

SNAPWrap v2.3.0 — Release Notes

What's New

Calibration Manager

A new **Calibration Manager** tool has been introduced. This provides a graphical interface for inspecting, repairing, and managing the calibration database directly from Mantid Workbench, without needing to edit files by hand.

Key capabilities:

- **Browse calibrations** — view all calibration states, versions, and their run ranges in one place, including which calibrations were propagated from another state and which run was used as the donor.
- **Propagate calibrations** — copy a calibration from one instrument state to another using a guided preview-and-execute workflow. A log of all propagation actions is kept automatically.
- **Detect and fix double-propagated entries** — the manager will flag calibration entries that have been propagated more than once and offer a repair option.
- **Delete corrupted entries** — a dedicated delete action is available for calibration versions that cannot be repaired automatically.
- **Repair a broken index** — if the calibration index file becomes corrupted or out of sync with the files on disk, SNAPWrap can now reconstruct it automatically from the version folders.

Cycle Date Awareness

SNAPWrap now tracks SNAP beam cycle dates and uses them when checking whether a calibration is valid for a given run. A calibration will only be applied if its cycle matches the run being reduced, preventing accidental use of out-of-date calibrations across cycle boundaries.

Materials Database

- A **Material Editor** dialog has been added, allowing users to add or edit entries in the materials database directly from the interface.
- **Quartz capillary** has been added as a built-in material option.

Sample Environment Equipment (SEE)

- Additional component attributes are now tracked for SEE assemblies, including a new collimator component.

- A new `extentScale` parameter has been added to `compositeBackground` for finer control over background region sizing.

Workspace Naming Improvements

- Output workspace names now correctly handle SNAPRed's native timestamp format.
- Workspace names with an optional suffix (text appended after the run number) are now recognised and processed correctly.
- Workspace names can now carry a sequence number on the prefix (e.g. `slice-001`, `slice-002`), making it possible to keep multiple named slices in the workspace tree simultaneously without them overwriting each other. Output workspaces from `resample` and `clean-tree` operations preserve the numbering automatically.

Bug Fixes

- **Q-space reduction with a calibration mask** — fixed a crash that occurred when requesting Q-space output while a calibration mask was active.
- **Q-space reduction with no difcal** — fixed a failure when both `qsp=True` and `continueNoDifcal=True` were set simultaneously.
- **filterLite with a mask** — fixed a crash in the lite-mode filter when a pixel mask was present.
- **Workspace naming after resample** — output workspace names were being constructed incorrectly; this has been corrected.
- **cleanTheTree timestamp handling** — the clean-tree operation was sometimes producing workspace names that still contained a timestamp; these are now always timestamp-free as intended. A follow-on fix also ensures correct behaviour for workspaces with numbered prefixes.
- **Calibration propagation** — several issues fixed: incorrect timestamp on propagated records, wrong run number used for cycle annotation, and version ordering bugs in `copyDifcal`.
- **Cycle dates file permissions** — SNAPWrap no longer crashes if the user does not have write access to the cycle dates file.
- **Exception handling** — improved error handling when loading state or calibration JSON files that are missing or malformed.